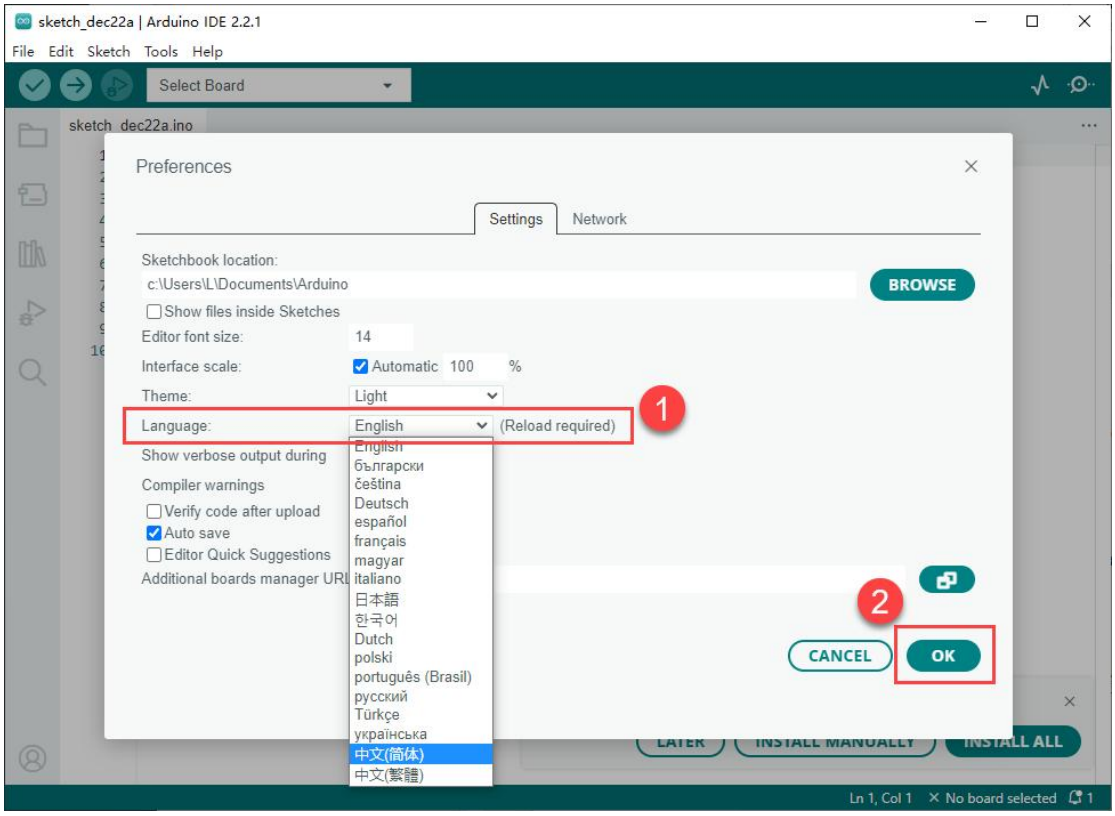
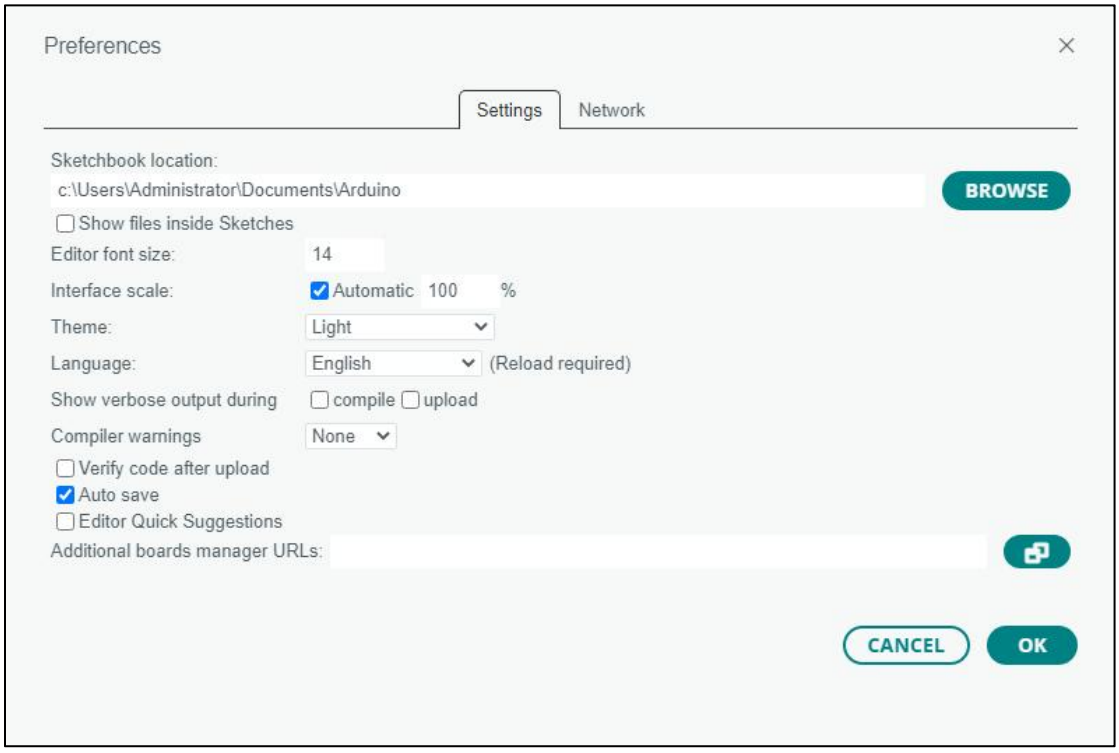
Arduino IDE Introduction

1. Arduino IDE Interface Setting

1) Modify the Chinese interface: select “File -> Preferences” on the Arduino IDE interface. Then, choose Chinese from the “Language”, and click “OK”.

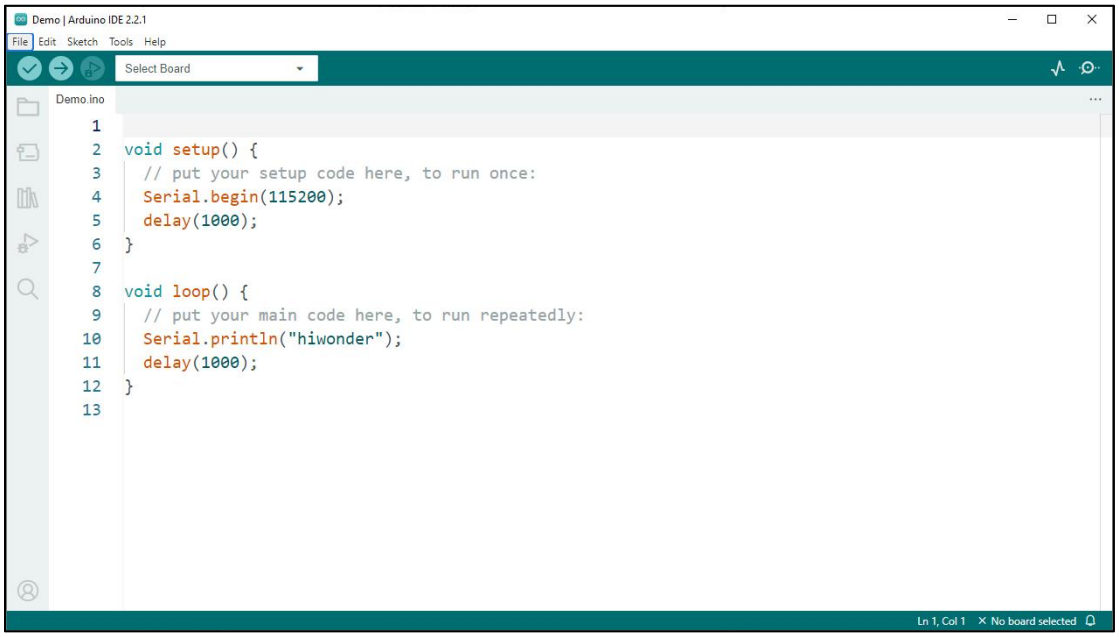


2) You can modify the project file path, editor text size, color themes, etc by selecting “File -> Preferences” in the pop-up window.

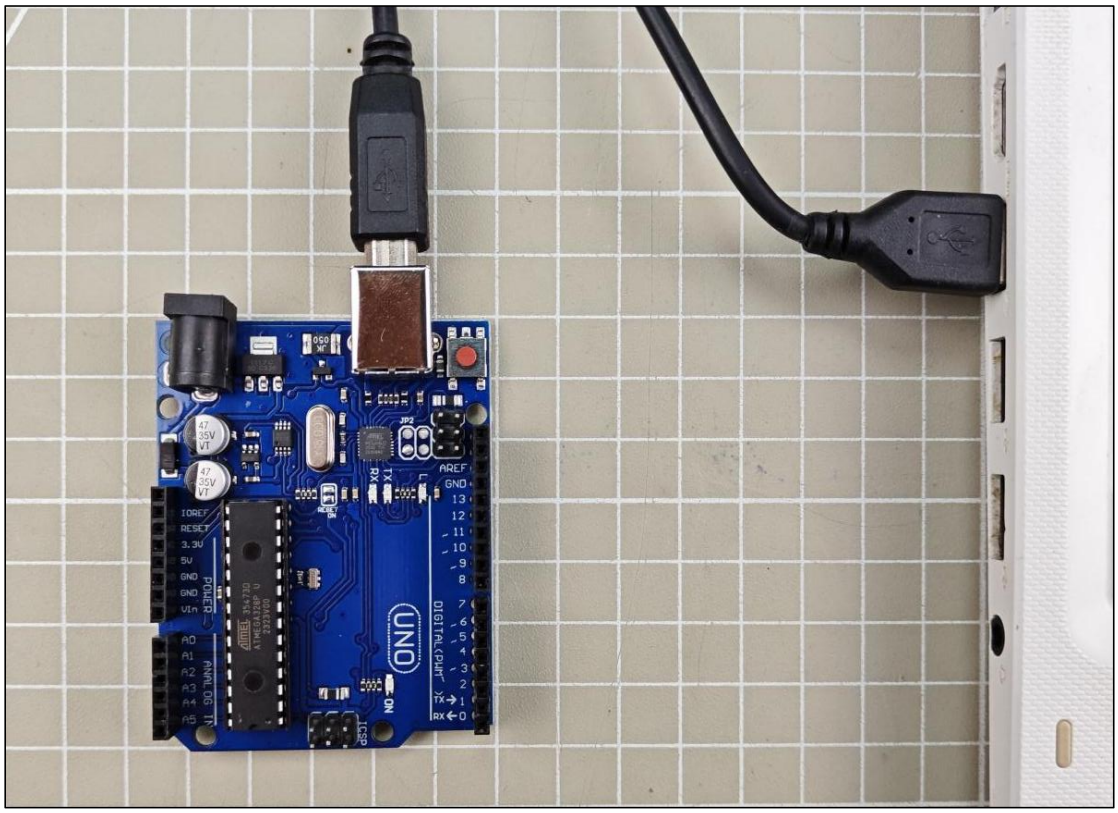


2. Arduino Download Program

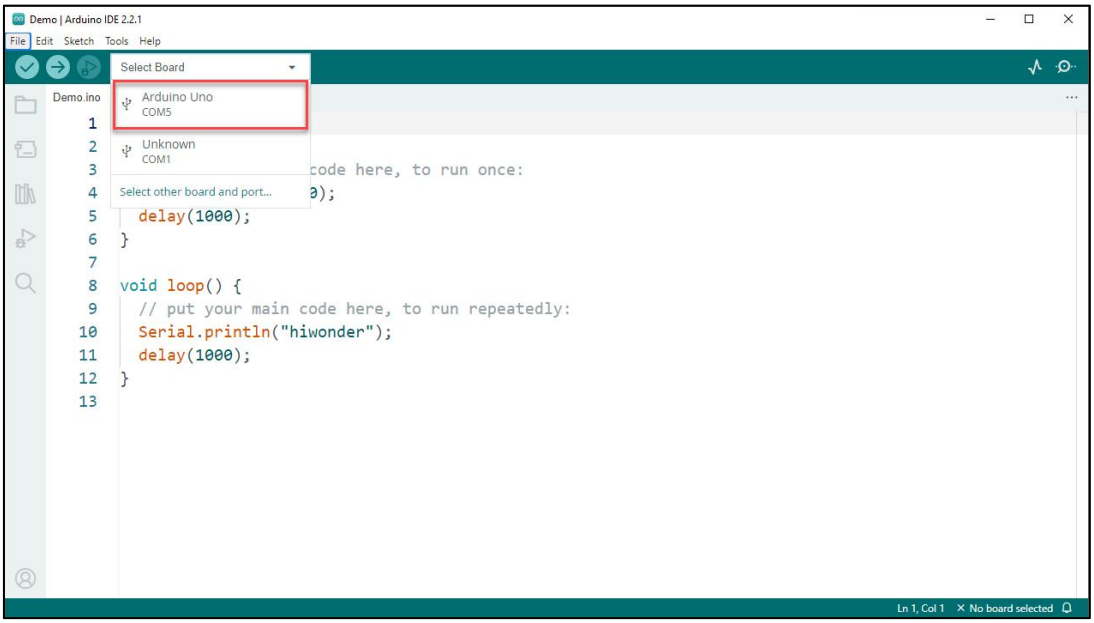
1) In this case, we will use an example that prints “Hiwonder” to illustrate. Open the example program “Demo.ino” in the “Demo” folder at the same folder as this document.



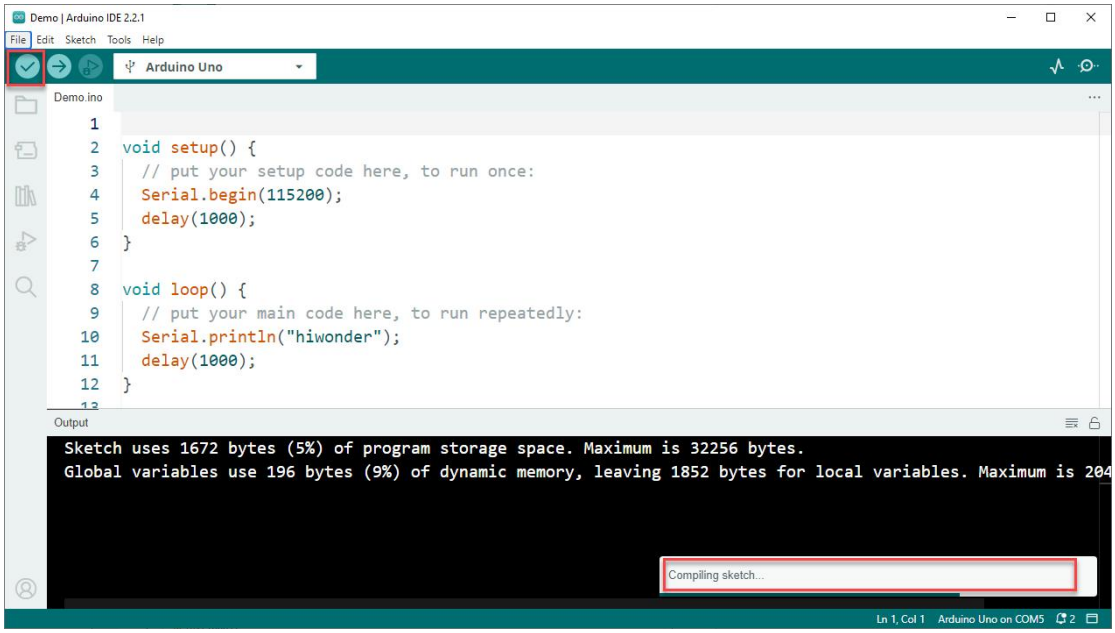
2) Connect Arduino board to the computer using Type-B cable.



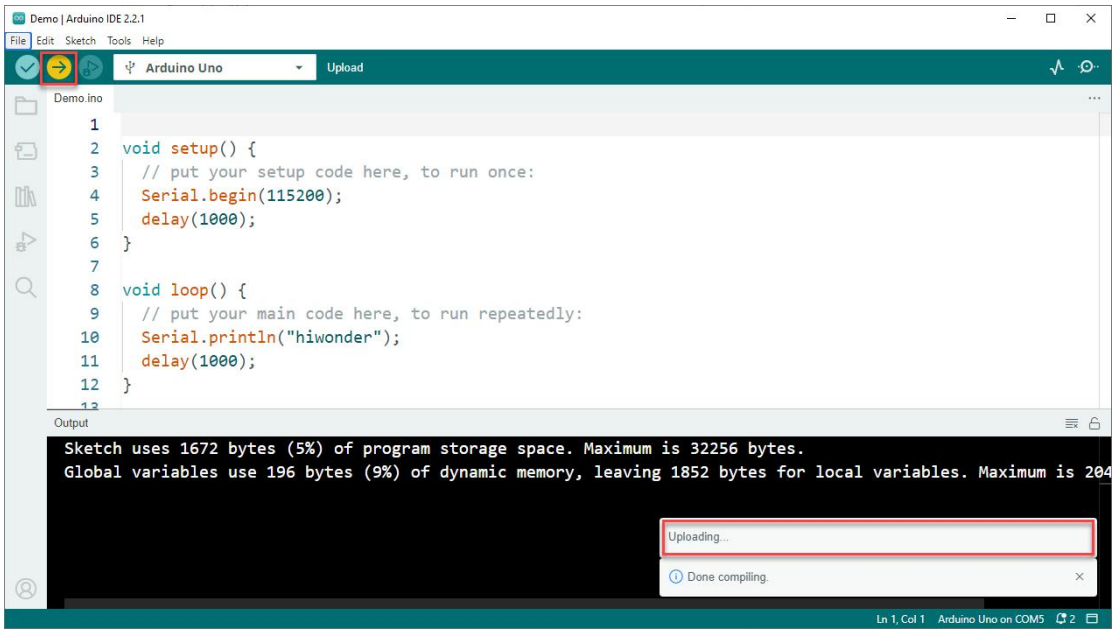
3) In the “Select board” option, find the corresponding Arduino development board. (Take “Arduino Uno” and “COM6” as en example to demonstrate. The COM port is not unique. You can can check the COM port number in the computer’s device manager.)



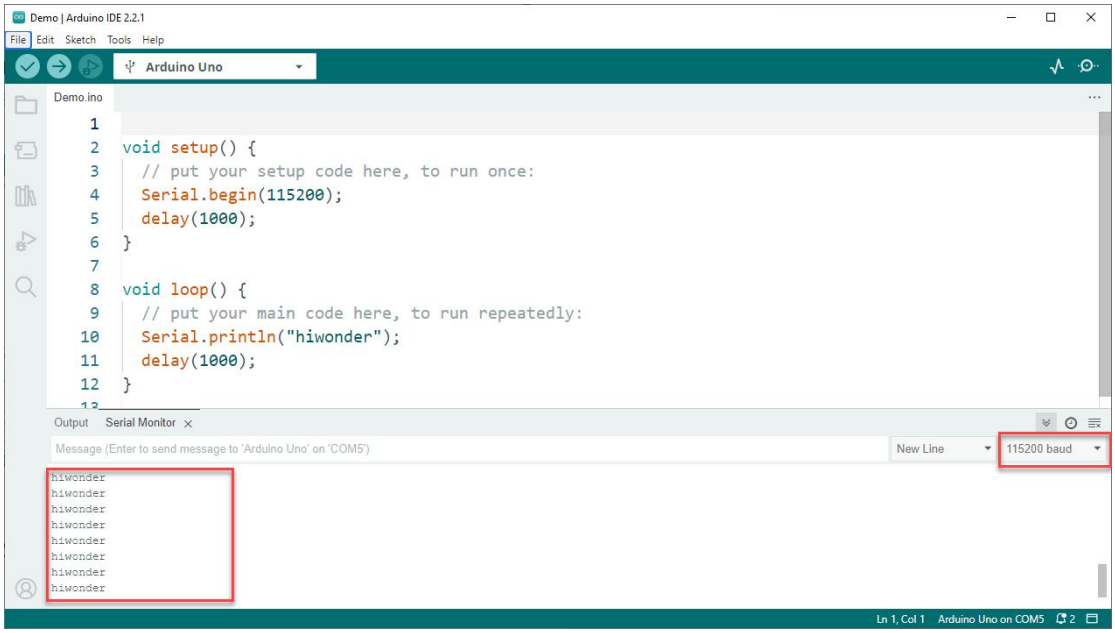
4) Click button to compile the program, which can check if the program has syntax errors and other issues.



5) After the compilation is complete, click to upload the program into the Arduino development board.



6) When the upload is done, click to open the serial monitor. The word “hiwonder” is printed on the serial monitor.



3. ESP32-Cam Environment Configuration

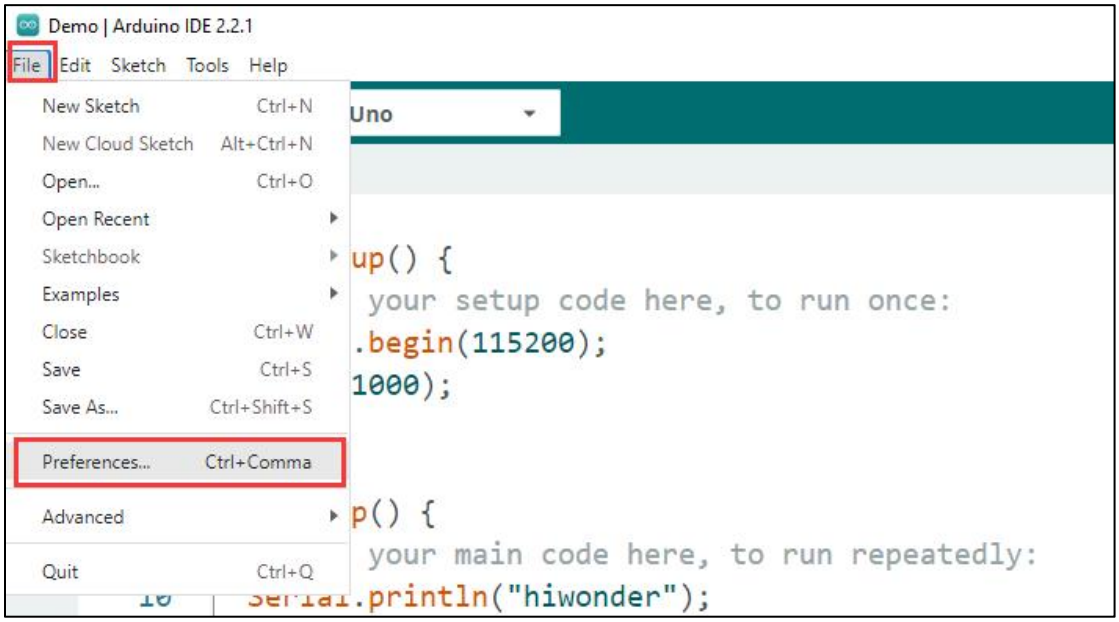
After the installation of programming software, configure this development board referring to text in the subsequent experience of ESP32-CAM. If not, related ESP32-Cam play program can not be download.

Two configuration methods are provided in this text. One is to directly install development board package with one click, which greatly improves the success rate of the installation. The other is to download the development board package on-line. By this method, users can choose any version of the package.

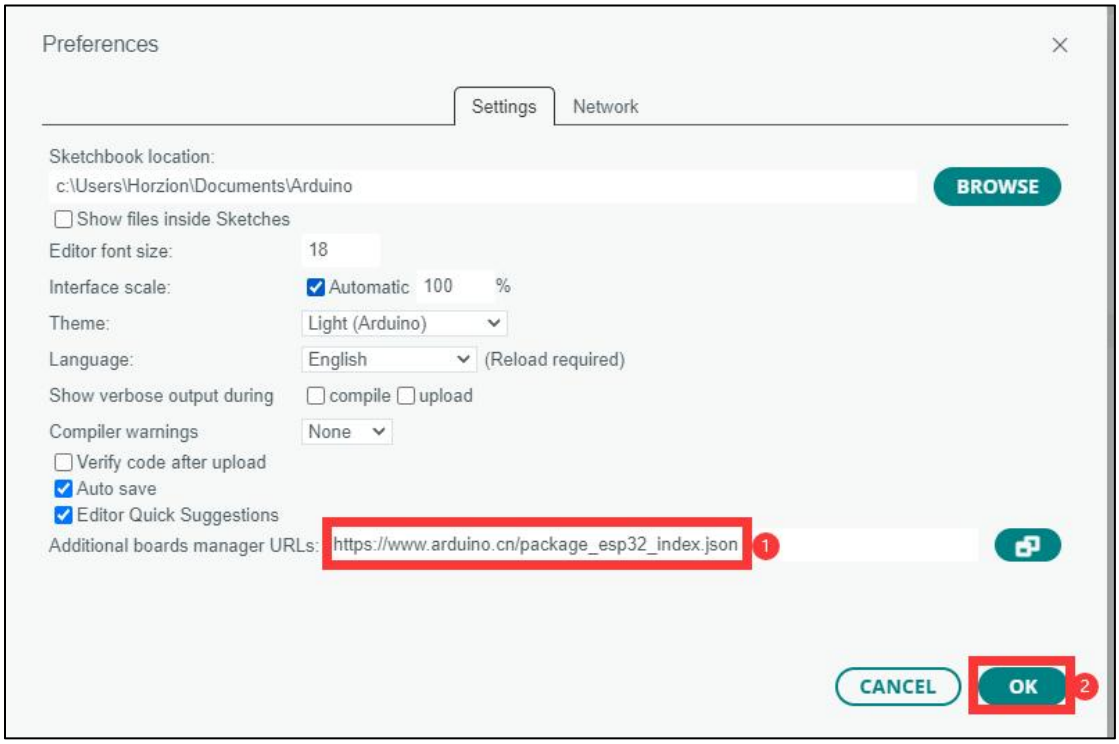
The following steps should be done with Internet.

3.1 One-Click Installation Development Board Package

1) Double click to open the edit interface, and then select “File-> Preferences” in turn.



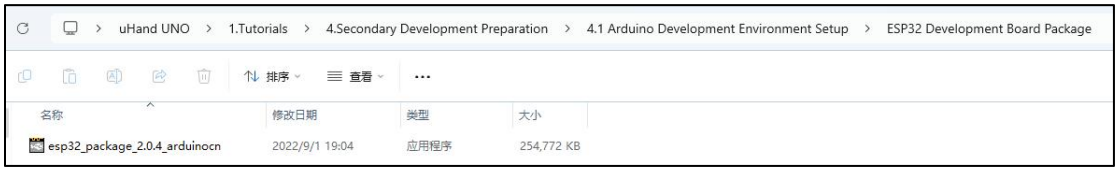
2) Find "Other Development Board Manager Address" box on the page. Copy the link: https://www. arduino. cn/ package _esp32 _index.json into it, and then click "OK".



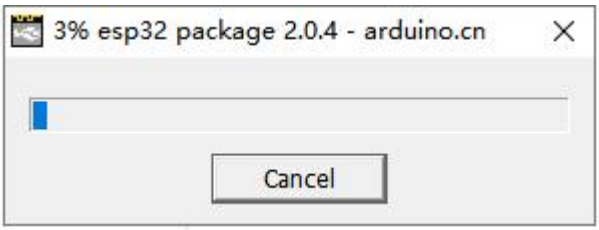
3) Close current software editing interface. (Do not skip this step or the configuration may fail.)

4) Find ESP32 Development Board package in the same path as in this

document.

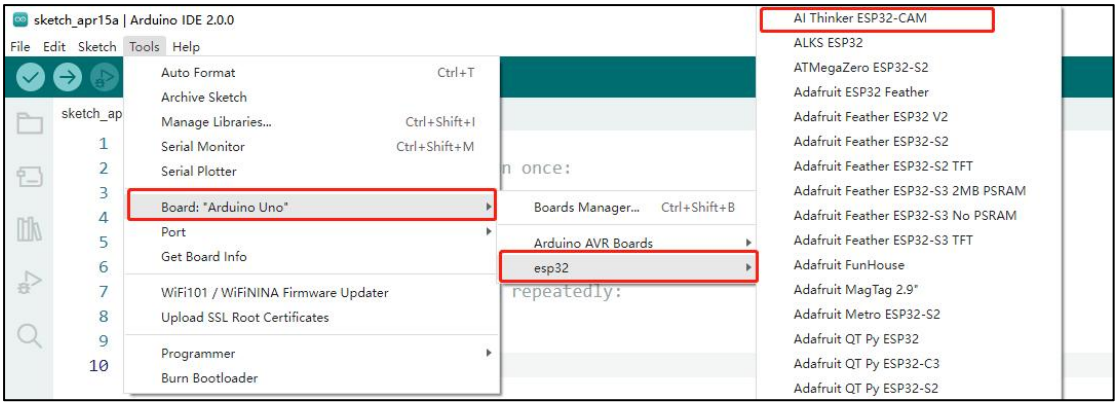


5) Then double click to install.



6) After successfully installing, double click to open the Arduino software. (If it was already open, you need to close and reopen it.)

7) To download ESP32-CAM related programme, click “Tool” and select “ESP32 Dev Module” in the development board options.

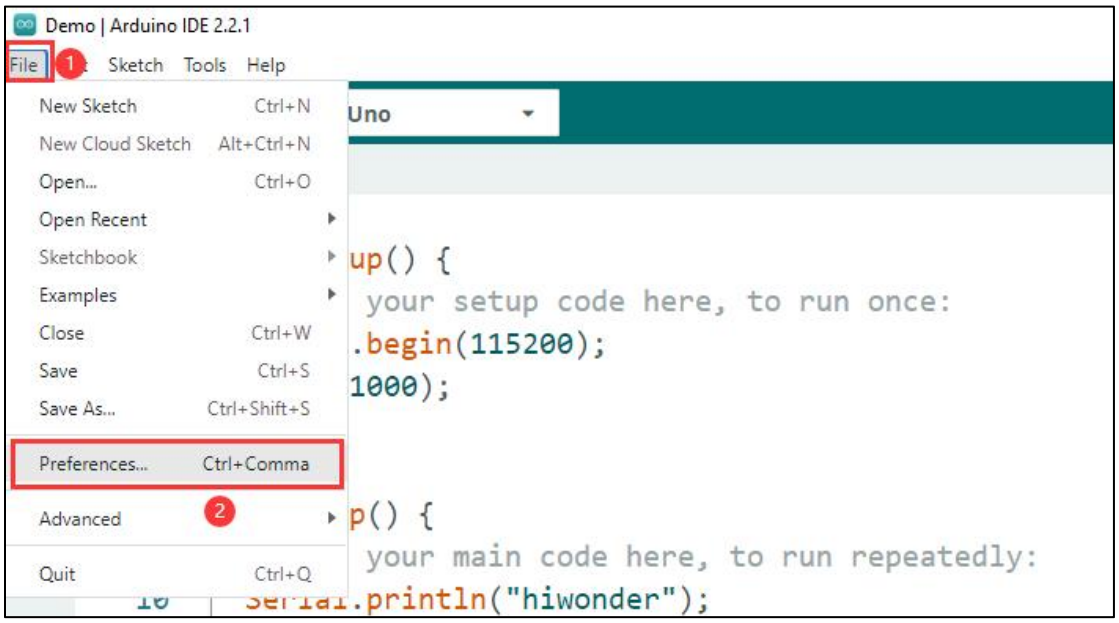


8) Here, the ESP32-CAM configuration is complete.

After the configuration is completed, if you can not find the ESP32 development board shown above, when you close and reopen the programming interface. Please repeat above steps.

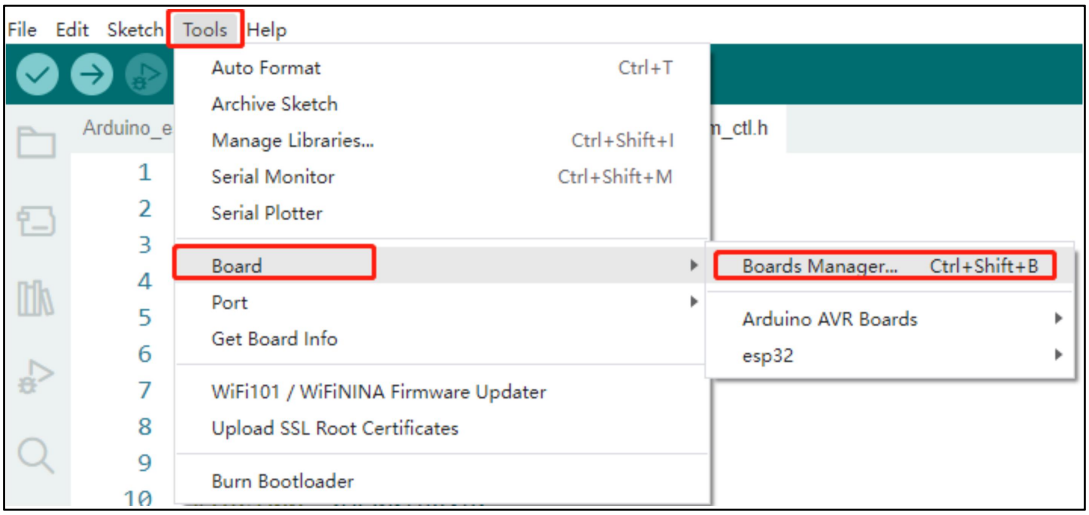
3.2 Download Development Board Package Online

1) Double click to open the programming software. Then select “File-> Preference” in turn.



2) Find "Other Development Board Manager Address" box on the page. Copy the link: https://www. arduino. cn/ package _esp32 _index.json into it. Then click "OK".

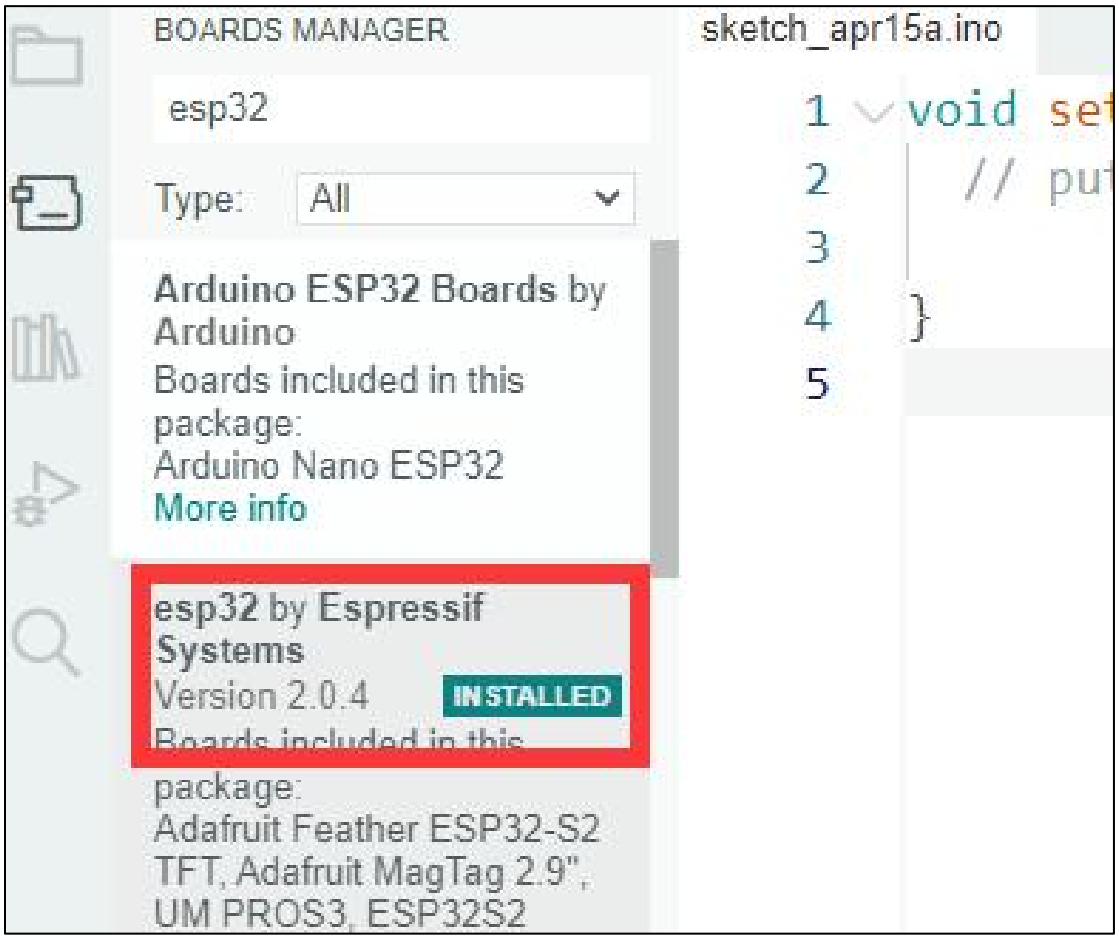
3) Then click "Tools-> Development Board-> Development Board manager" in turn.



4) In the version selection box, go down to the ESP32 Development Board. Click "Select Version" drop-down button, and select a development board package version (Preferable version is 1.0.5 above, in this case version 2.0.4.), and click "Install".

5) After the installation is complete, click "Close". (The installation

process takes some time, please be patient. If the installation fails, you can try again by clicking “Install”.)



4. Import Library File（Optional）

If the program requires importing the library files, you can refer to the following operations:

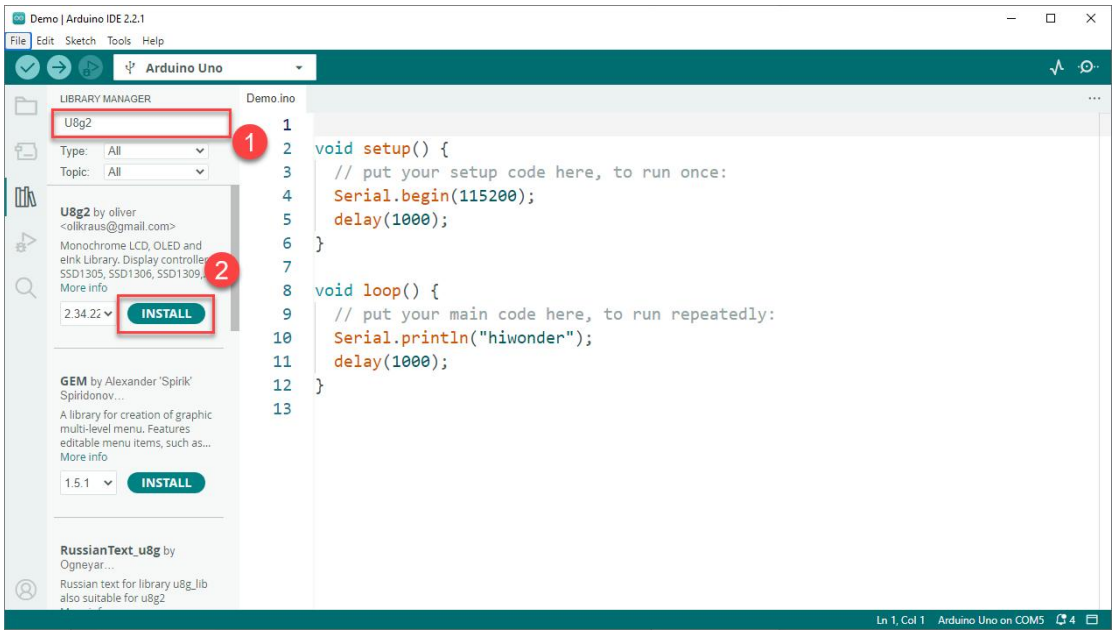
1) Online Importing

This method is usually used for importing some official Arduino library files, for example, the library file for driving LED dot matrix (TM1640), the library file for OLED screen module (U8g2).

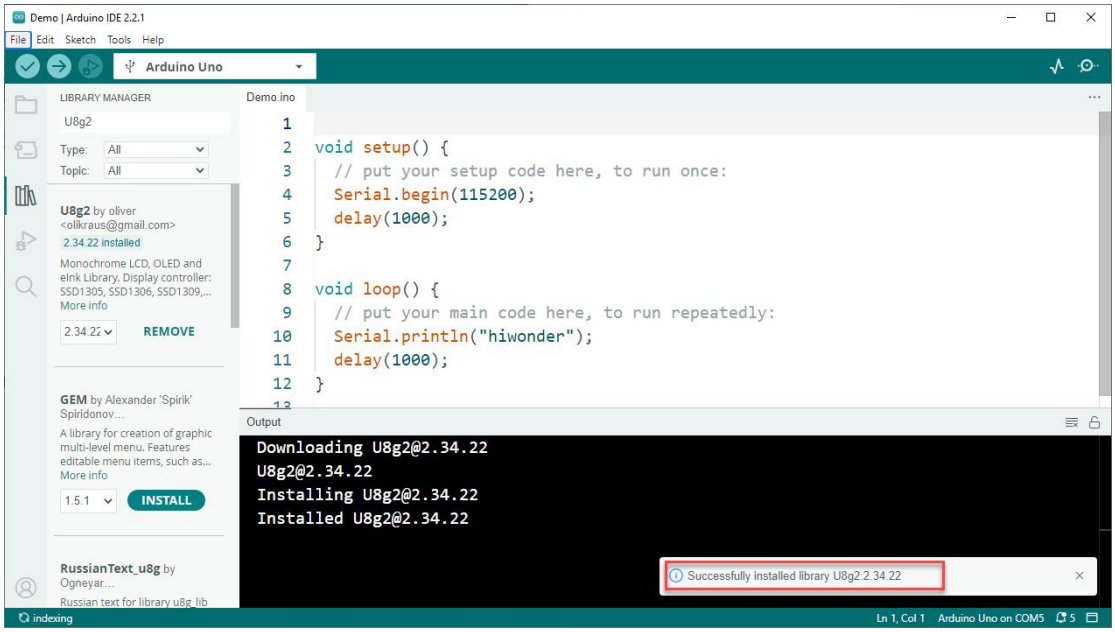
Take the library file for driving LED dot matrix (TM1640) as an example to illustrate:

① Click at the sidebar of Arduino IDE interface.

② In the pop-up library management bar, enter “U8g2” to automatically search for the library files, then click “install”.



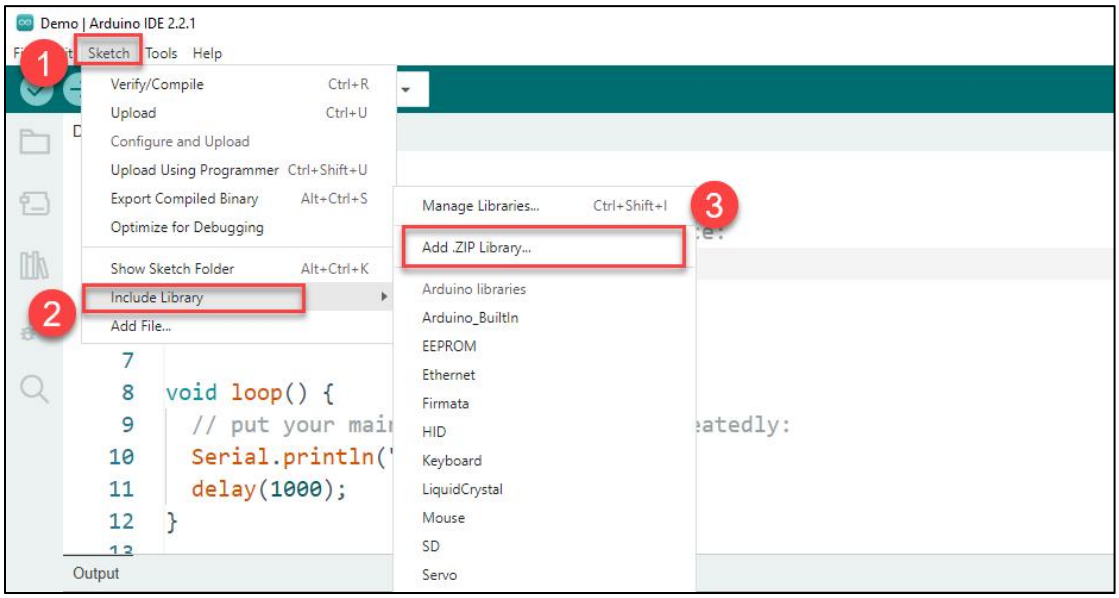
③ The prompt shown in the red box indicates the successful installation.



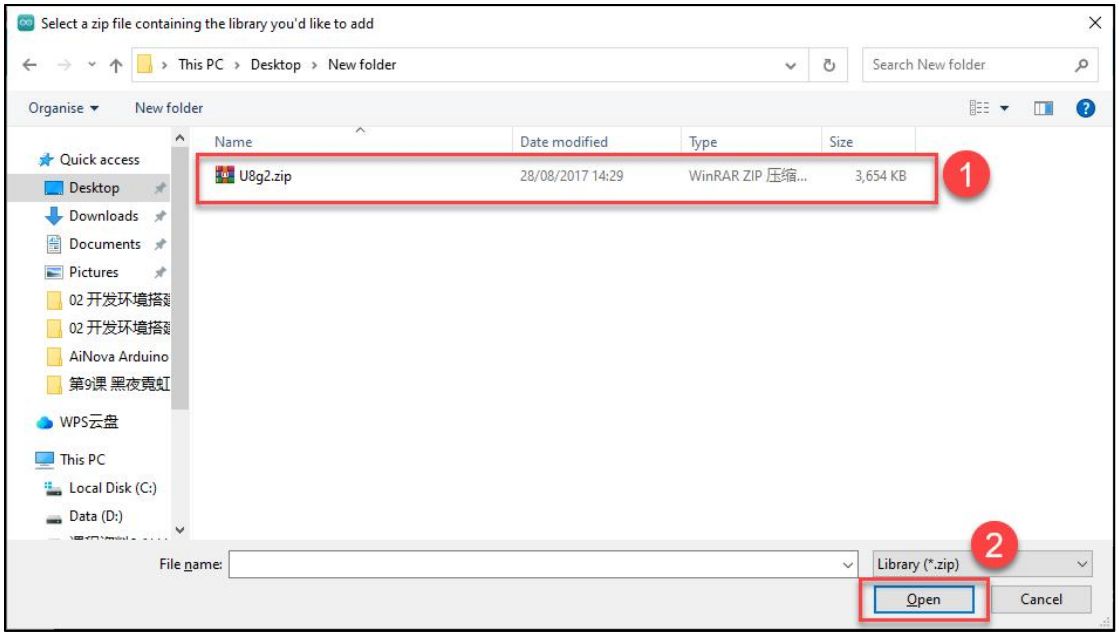
2) Local Importing:

This method is usually used for downloading library file or customizing the library file importing. Here will take the “U8g2” library file as an example to illustrate.

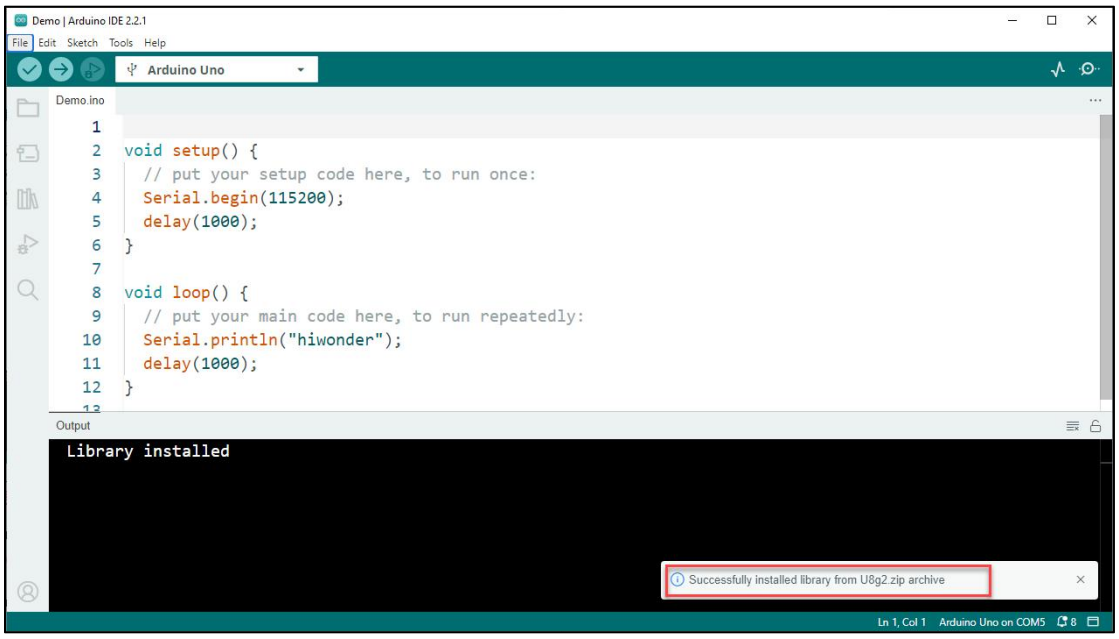
① Select “Sketch -> Import library-> Add .ZIP library”.



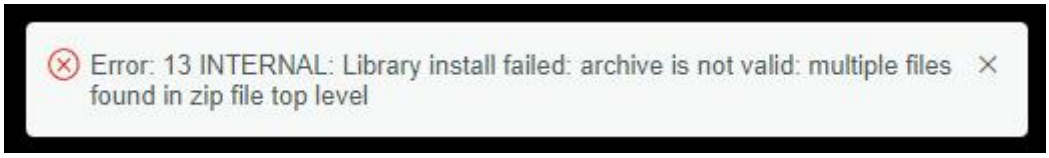
② Search for the U2g2.zip and open it.



③ The prompt shown in the below red box indicates that the library file has been installed already.

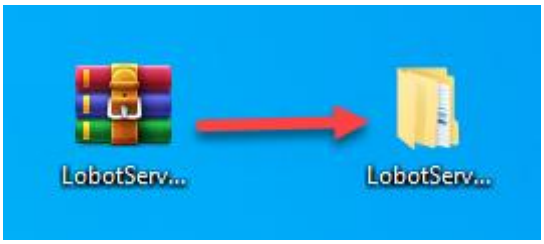


! In the process of importing the library file in the local, if you encounter the below error during the process of importing library files locally, you can refer to the following solution.

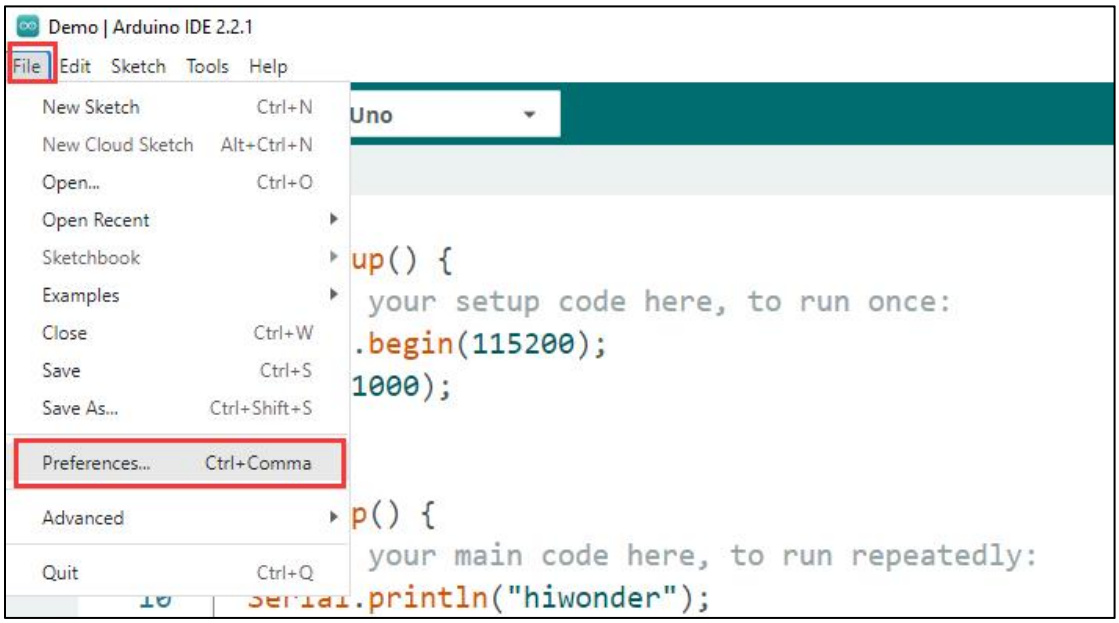


Take the library file “LobotServoController” as example to illustrate. In Arduino IDE versions 1.8.5 and above, the error above may arise when importing this library file. You can solve this problem by extracting the library file and placing it directly into the 'libraries' folder in the path of the project. The specific solution is as follows:

- At first, extract the library file “LobotServoController”.

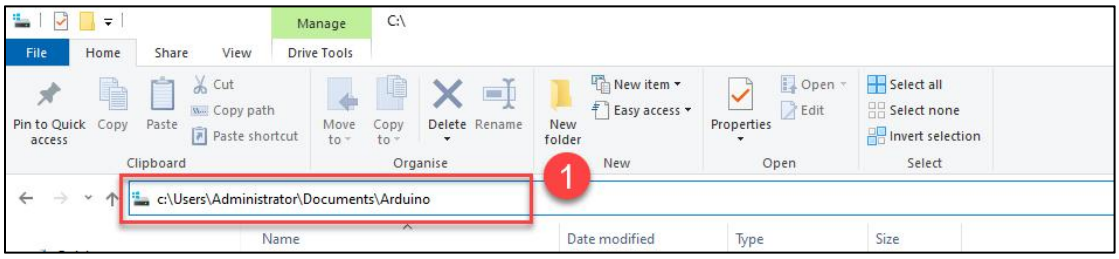


- Click “File->Preferences” in sequence.

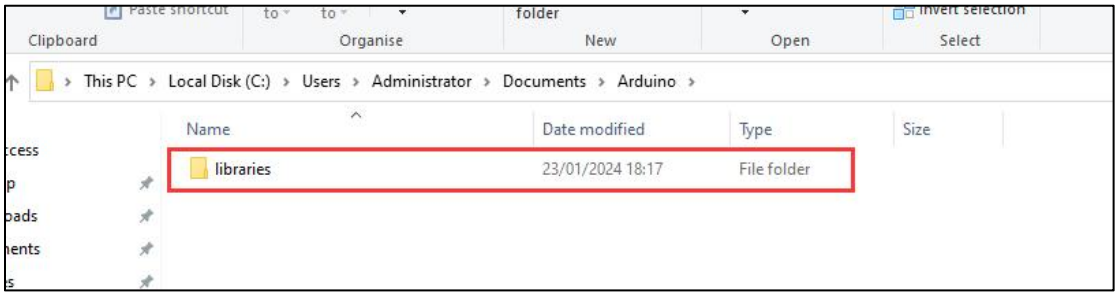


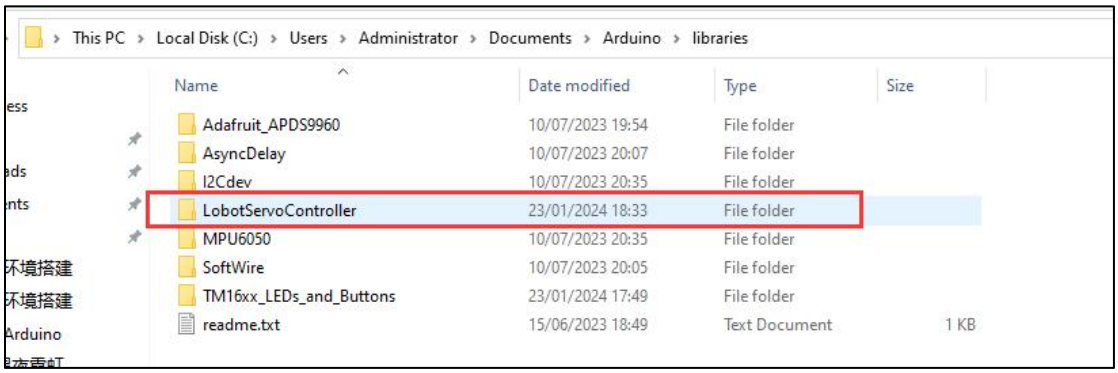
- In the opening preferences, copy the path.

- Paste it in my computer, and then press “Enter” to access the path.



- Place the previously extracted library file into the “libraries” folder.





- Restart Arduino IDE. The library file has been located in the library of Arduino IDE.